

SIMATS ENGINEERING

ASSESSMENT TOOL 3

COURSE NAME: Operating Systems

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COURSE CODE: CSA-0409

REGNO:192511401

COURSE OUTCOME COVERED

CO3: Evaluate memory management mechanisms such as linking, paging, and segmentation to determine their effectiveness for different system requirements

Assessment Tool Weightage: 30%

QUESTIONS: 25

Total Marks: 50

MCQ TEST:

1. In a multiprogramming system, a process generates logical addresses which are translated into physical addresses before accessing memory. Which of the following correctly describes the relationship between logical and physical addresses?
 - a) Logical address is directly used by memory
 - b) Physical address is generated by CPU
 - c) Logical address is translated into physical address by MMU
 - d) Both are same

Answer: c

2. A system uses base and limit registers for memory protection. If a process generates a logical address beyond the limit value, what will the system most likely do?

- a) Adjust the address automatically
- b) Allow access
- c) Generate a trap (error)
- d) Ignore the address

Answer: c

3. In a paging system, the logical address consists of a page number and an offset. If the page size is fixed, what is the main advantage of using paging?

- a) Eliminates internal fragmentation

- b) Eliminates external fragmentation**
- c) Reduces page faults**
- d) Increases memory size**

Answer: b

4. A process accesses memory frequently, and each access requires two memory references (one for page table, one for data). Which hardware component is used to reduce this overhead and how?

- a) Cache memory by storing instructions**
- b) TLB by storing recent page table entries**
- c) Registers by storing pages**
- d) Disk by storing data**

Answer: b

5. In a system with a TLB, when a logical address is generated, the system first checks the TLB. If the required entry is not found, what sequence of operations will occur?

- a) Directly access physical memory**
- b) Search page table, update TLB, then access memory**
- c) Restart process**
- d) Ignore request**

Answer: b

6. Consider a system where each page table entry contains frame number and control bits. Which of the following is NOT typically included in a page table entry?

- a) Frame number**
- b) Valid/invalid bit**
- c) Protection bit**
- d) CPU register value**

Answer: d

7. A system uses segmentation where programs are divided into logical units such as code, data, and stack. What is the primary advantage of segmentation compared to paging?

- a) Fixed size memory blocks**
- b) Logical program structure representation**
- c) Faster access**
- d) No fragmentation**

Answer: b

8. In segmentation, each logical address consists of a segment number and an offset. If the offset exceeds the segment limit, what will happen?

- a) Address wraps around**
- b) Process continues normally**
- c) Trap is generated**
- d) Segment is expanded**

Answer: c

9. Which of the following best describes the role of protection bits in both paging and segmentation systems?

- a) Increase memory size**
- b) Control access permissions like read/write**
- c) Store address values**
- d) Replace pages**

Answer: b

10. A system implements virtual memory using demand paging. When a page fault occurs, which of the following steps is performed first?

- a) Replace a page**
- b) Check if reference is valid**
- c) Load page from disk**
- d) Restart instruction**

Answer: b

11. In a virtual memory system, if no free frame is available during a page fault, which mechanism is used to select a page for removal?

- a) Memory allocation**
- b) Page replacement algorithm**
- c) TLB**
- d) Segmentation**

Answer: b

12. Consider FIFO page replacement. If pages are loaded in order 1, 2, 3, and a new page must be loaded, which page will be removed?

- a) Page 3
- b) Page 2
- c) Page 1
- d) Random page

Answer: c

13. In LRU page replacement, the system keeps track of page usage. Which page is replaced when memory is full?

- a) Most recently used
- b) Least recently used
- c) First loaded
- d) Random

Answer: b

14. Which of the following scenarios best demonstrates internal fragmentation in paging?

- a) Variable memory blocks
- b) Unused space within a fixed-size page
- c) Unallocated segments
- d) Missing pages

Answer: b

15. Which of the following scenarios best demonstrates external fragmentation?

- a) Fixed-size pages
- b) Scattered free memory blocks
- c) Full memory
- d) Page fault

Answer: b

16. In linking, multiple object modules are combined into a single executable. What is the key difference between static and dynamic linking?

- a) Static linking at runtime
- b) Dynamic linking before execution
- c) Static linking includes libraries at compile time
- d) Both are same

Answer: c

17. During dynamic linking, when is the required library code loaded into memory?

- a) Compile time**
- b) Load time**
- c) Run time**
- d) Before compilation**

Answer: c

18. Which of the following best describes the role of a loader in memory management?

- a) Compiles program**
- b) Executes program**
- c) Loads program into memory for execution**
- d) Creates page table**

Answer: c

19. A system uses both paging and segmentation. Which of the following is true about this combined approach?

- a) Only paging is used**
- b) Only segmentation is used**
- c) Segments are divided into pages**
- d) Pages are divided into segments**

Answer: c

20. In a TLB-based system, effective memory access time depends on hit ratio. If hit ratio increases, what happens?

- a) Access time increases**
- b) Access time decreases**
- c) No change**
- d) System slows down**

Answer: b

21. Which of the following conditions leads to a page fault?

- a) Page present in memory**
- b) Page not present in memory**
- c) TLB hit**
- d) Segment valid**

Answer: b

22. What is the main purpose of virtual memory in modern operating systems?

- a) Increase CPU speed**
- b) Allow execution of large programs**
- c) Reduce disk usage**
- d) Eliminate RAM**

Answer: b

23. In address binding, if addresses are bound during execution time, what feature is required?

- a) Static loading**
- b) Dynamic relocation**
- c) Paging**
- d) Segmentation**

Answer: b

24.

Which of the following best describes demand paging?

- a) Load all pages at once**
- b) Load pages when required**
- c) Remove pages**
- d) Increase page size**

Answer: b

25.

In a system with heavy multiprogramming, why is memory protection important?

- a) Increase speed**
- b) Prevent processes from interfering with each other**
- c) Increase memory size**
- d) Reduce page faults**

Answer: b

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary (5 Marks)	Level 3 – Proficient (4 Marks)	Level 2 – Developing (2–3 Marks)	Level 1 – Beginning (0–1 Mark)
Conceptual Understanding	30%	Demonstrates thorough understanding; answers all questions accurately	Shows good understanding with few errors	Partial understanding; several incorrect responses	Lacks understanding; majority of answers incorrect
Accuracy of Answers	25%	All answers are correct	Most answers are correct with minor mistakes	Some correct answers; frequent errors	Mostly incorrect answers
Application of Knowledge	20%	Correctly applies concepts to problem-based questions	Applies concepts with minor errors	Limited ability to apply concepts	Unable to apply concepts
Time Management	15%	Completes quiz well within time efficiently	Completes on time with minor delays	Requires extra time or rushes through questions	Unable to complete within given time
Question Interpretation	10%	Interprets all questions correctly and responds appropriately	Misinterprets a few questions	Often misinterprets questions	Unable to understand questions